

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$B = (\vec{v}_1, \vec{v}_2) = \left( \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$$

$$\left( \begin{array}{cc|cc} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 7 \end{array} \right) \hat{=} \left( \begin{array}{cc|cc} 7 & -3 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{array} \right)$$

$$[X]_B = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix}$$

$$[y]_B = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix}$$

$$[x]_B^T \cdot [y]_B = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix}^T \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix} =$$

$$= 49x_1y_1 - 21x_1y_2 - 21x_2y_1 + 9x_2y_2 +$$

$$+ 4x_1y_1 - 2x_1y_2 - 2y_2x_1 + x_2y_2 =$$

$$= 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 10x_2y_2$$